HybridKernel: A Profiler-Driven Boundary-Fusion Gate for Hybrid Attention/SSM Serving

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Abstract

Hybrid attention/state-space language models invite a systems question: do attention-to-SSM layer boundaries create avoidable conversion, materialization, launch, or locality overhead that a fused boundary operator could remove? We report a Mac-feasible pre-GPU gate for this question. Architecture maps show large boundary-crossing activation streams, but a runtime audit weakens the broad novelty claim because current vLLM hybrid SSM support already handles important state-layout and transfer paths. A threshold model says Granite 4.0 H needs roughly 25% genuinely avoidable boundary traffic at 60% recovery to clear a 3% proxy gain; Qwen3-Next needs 10.4% but is less directly matched to the Granite Mamba2 hypothesis. We therefore do not claim a kernel speedup. The contribution is a reproducible profiler gate, a fixed-request vLLM driver, a native packet skeleton generator, a pre-registered profiler-analysis parser, and an artifact-completeness verifier that rejects client-only traces, unfilled skeletons, and stale analysis before any native result can be cited.

1 Hypothesis

HybridKernel tests whether adjacent attention and SSM layers expose boundary-local overhead not already removed by serving systems. Candidate costs include layout conversion, residual-stream materialization, host launch gaps, cache-locality loss, or redundant memory traffic. The hypothesis is deliberately narrow: a useful kernel must compose with existing vLLM hybrid mechanisms rather than replace them.

2 Current Evidence

3 Profiler Gate

The next experiment is a native NVIDIA/vLLM run with Nsight Systems and Nsight Compute. It must show a distinct boundary-local overhead attributable to attention/SSM transitions, not warmup, graph capture, batching, or unrelated kernels. The runbook now specifies exactly how to reduce Nsight traces into parser fields: total step time, boundary-local time, matched non-boundary control time, and recoverable fraction. Promotion requires at least three repeated runs whose recoverable-gain upper bound clears 3%. The analysis parser computes

$$\text{gain}_{ub} = \frac{\max(\text{boundary ms} - \text{matched control ms}, 0)}{\text{total step ms}} \times \text{recoverable fraction.}$$

If repeated native summaries show less than 1% recoverable gain, the branch should be killed or shelved. Before any run is cited, the artifact verifier must also pass on the exact run directory, confirming environment metadata, server-side Nsight Systems and Nsight Compute scope, profiler artifacts, profiling logs, the pre-registered readout questions, at least three distinct repeated metric rows for one model, and profiler-analysis outputs that match the same metric file. A packet generator creates this directory shape for GPU operators, but the checker rejects its `TODO_NATIVE_PROFILE_FILL` sentinels until real metadata,

Artifact	Result	Decision
Architecture map	Granite has 8 boundaries and 20.0% boundary stream fraction; Qwen3-Next has 23 inferred boundaries and 47.9%.	Inspectable, not speed evidence.
Runtime audit	vLLM already handles hybrid state layout and transfer paths.	Broad novelty weakened.
Threshold model	Granite needs about 25.0% avoidable boundary traffic at 60% recovery for a 3% proxy gain.	Mac kernel work not justified.
Profiler driver	Fixed-request OpenAI-compatible driver dry-runs locally; runbook now profiles the vLLM server, not just the HTTP client.	Native gate is less likely to be mis-run.
Analysis parser	JSON profiler summaries are reduced into avoidable-boundary share and recoverable-gain upper bound.	Decision rule is pre-registered.
Artifact verifier	Native run directories are checked for metadata, server-side Nsight Systems and Compute scope, Nsight artifacts, logs, readout rows, distinct repeated metric rows, and matching profiler-analysis outputs.	Client-only, duplicated-trace, and stale-analysis evidence is rejected.
Packet generator	A local script creates the required native run-packet skeleton with TODO sentinels.	GPU operators get a consistent handoff; skeletons are rejected as evidence until filled.
Synthetic packet fixture	A Mac-local fixture documents the packet shape and exercises the checker with placeholder Nsight files.	Fixture is not profiler evidence.
Packet checklist	The native handoff now has a single required packet layout and stop-local-work decision.	Local readiness is saturated until GPU data exists.

Table 1: HybridKernel evidence before native GPU profiling.

metrics, and readout entries replace them. This is an admissibility check only; it does not promote the method.

4 Claim Boundary

Allowed now: HybridKernel is a weakly alive profiler-driven systems branch with a runnable native measurement plan. Not allowed: throughput, latency, HBM, energy, occupancy, or vLLM speedup claims. If native profiling does not reveal separable boundary overhead, the branch should be killed.

5 Artifacts

- `experimental/hybridkernel/phase2/nvidia_vllm_profiler_runbook.md`
- `experimental/hybridkernel/phase2/profiler_driver.py`
- `experimental/hybridkernel/phase2/create_native_run_packet.py`
- `experimental/hybridkernel/phase2/analyze_profiler_metrics.py`
- `experimental/hybridkernel/phase2/check_profiler_run_artifacts.py`
- `experimental/hybridkernel/phase2/native_run_packet_checklist.md`
- `experimental/hybridkernel/phase2/profiler_analysis_gate.md`

- experimental/hybridkernel/phase2/pre_gpu_threshold_model.md
- experimental/hybridkernel/paper/colm_workshop_scaffold.md